

AMARNATH S. PATEL

Orlando, FL | amarnath.patel@ucf.edu | apatel.co | github.com/jeebuscrossaint

Physics and mathematics undergraduate — numerical methods, statistical inference, and performance-critical C++/CUDA.

EDUCATION

University of Central Florida **Undergraduate Student** **4.00 GPA**
B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor August 2025 – Expected May 2029
• **Mathematics:** Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Applied Linear Algebra, Ordinary Differential Equations, Mathematical Methods for Physics, Discrete Structures, Honors Calculus III
• **Computing:** Data Structures & Algorithms, Computer Architecture, Computer Logic Design, Object-Oriented Programming, Deep Learning

Florida Atlantic University **University coursework via FAU High School, ages 14–18** **3.66 GPA**
111 credit hours completed between ages 14 and 18, alongside high school August 2021 – May 2025

QUANTITATIVE RESEARCH EXPERIENCE

Independent Research — [CELERIS](#): high-performance numerical solver. C++23, CUDA, Python 2026
• Wrote a numerical solver for Maxwell's equations in periodic structures from scratch: Fourier-basis eigenvalue problems per layer, dense complex linear algebra, and a numerically stable scattering-matrix recursion chosen over the transfer-matrix formulation precisely because the latter loses conditioning as layers accumulate.
• Held the whole stack to measurable error bounds rather than eyeballed agreement — $1e-6$ against an independent method, $1e-15$ on multilayer recursion, $5e-12$ on a degenerate-case reduction, unitarity to 1.000000 — and cross-validated the finished solver against two third-party implementations to $\approx 1e-7$.
• Gradient-based (Adam) optimization over high-dimensional design parameters, plus Monte-Carlo tolerance analysis quantifying how performance degrades under manufacturing noise.
• Wrote the hot path as a CUDA kernel and benchmarked it against the project's own optimized 16-core CPU implementation: $4.8\text{--}5.8\times$ at 92k elements, agreeing to machine precision. Read the speedup's decline with problem size as evidence the kernel was memory-bound rather than compute-bound. *Sole-author manuscript in preparation.*

Undergraduate Researcher — [FAU Grant-Funded AI Safety Research Project](#) January 2024 – March 2025
Supervisor: Tucker Hindle, Florida Atlantic University
• Ran a signal-versus-noise study on a 255,000-passage corpus: benchmarked 10 candidate statistical discriminators (perplexity, transformer classifiers, latent semantic analysis, natural-language inference, burstiness) and ranked them by separation power. GPT-2 perplexity won at $3.35\times$ separation (17.5 versus 58.5); BERT next-sentence prediction turned out to carry no signal at all, which was worth establishing.
• Built four adversarial strategies against those discriminators — iterative rewriting, tree-search decoding, list-branching and reinforcement learning — over 3,125 generated sequences, and characterized a consistent tradeoff between evasion success and output quality. Grant-funded with HPC access.

Undergraduate Researcher — [UCF Astrophotonics Lab](#), CREOL August 2025 – Present
Supervisor: Dr. Stephen Eikenberry
• Estimation and signal processing on noisy experimental data: FFT sideband isolation and demodulation, Butterworth filtering, and a joint nonlinear optimization over three coupled parameters to recover complex amplitude and phase from raw interferograms — $\approx 98\%$ fidelity against simulated ground truth.
• Automated the four-instrument acquisition chain end to end, with in-loop optimization of a live objective and automatic rejection of bad frames, turning days of manual data collection into a single unattended run.

PROGRAMMING

• **Systems, in C, Rust and Go:** [hush](#), an interactive shell in C; [limebar](#), a Wayland status bar in C; [constrict](#), a build system in Go written to replace make; [hazmat](#), a password manager in C99.
• **Problem solving:** [Project Euler](#) and multi-year [Advent of Code](#) solutions in C++, plus [aocli](#), a command-line client for the latter. 1st place, Night Hacks hackathon.

SKILLS

Mathematics and methods: linear algebra and eigenvalue problems, complex analysis, differential equations, Fourier analysis, numerical optimization, Monte-Carlo methods, statistical benchmarking and discriminator evaluation, error analysis

Programming: C++ (C++23), Python (NumPy, SciPy), C, CUDA, Rust, Go; Linux, Git, CMake, GPU profiling

HONORS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) 2025
Lockheed Martin Award, "Highest Level of Engineering Excellence," AEV Competition 2024
1st Place, Night Hacks Hackathon — UniUtils (course schedule generator and classroom finder) 2023